Exercises: Decision Networks and Heuristic Decision-Making

Goals

1. *Understand graphical models and Decision Networks*
2. *Apply Heuristic Rules*

Exercises

1. Recall the exercises 3-to-8 from Lab 1. All of them present Decision-Making problems and you were asked to solve them by using different Utility Theory models. Now, it is the time to represent each of these problems graphically with a Decision-Network.

2. The Sprinkler problem revisited. Solve this new situation using Decision Networks:

The Sprinkler problem with different data:

Its summertime and you have to decide to turn on your sprinklers or not. ***It is important to keep your grass wet as the high temperatures during the day may ruin your grass.*** But the decision depends on the weather: if it rains, there is a high probability that the grass will be wet, so the sprinklers would not be necessary. Considering the following probability tables:

Rain

Rain	P(Rain)
T	0,4
F	0,6

GrassWet

Rain	P(GrassWet="T"|Rain)
T	0,7
F	0,3

¿Would you turn on the sprinklers or not?

3. The Bet problem revisited. Will you accept the bet with this new formulation? Solve this new situation using Decision Networks.

The Bet Problem: John is offering you a bet. If your favourite team, the "Obra", wins, John is inviting you to go out for dinner, otherwise you will invite him. You spend around 20 euros for this type of dinner. When deciding whether to accept this bet, you will have to assess her team's chances of winning (which will vary according to the humidity of the day). You also know that you will be happy if your team wins and miserable if it loses, regardless of the bet.

Result

Result	P(Result)
Owins	0,8
Olooses	0,2

Humidity

Result	Humidity	P(Humidity|Result)
Owins	Dry	0,6
Owins	Wet	0,4
Olooses	Dry	0,4
Olooses	Wet	0,6

Utility

Bet	Result	U(Result,Bet)
YesBet	Owins	10
NoBet	Owins	6
YesBet	Olooses	-10
NoBet	Olooses	-5

¿Are you accepting the Bet if the Humidity Forecast is "Wet"?

4. Solve both the Sprinkler and the Bet problem by means of Heuristic Rules.

5. Moral dilemmas. Consider these situations and decide **on each step** if the action is acceptable or not (Alexandra Gomis, 2016):

- Dilemma 1:

    1. The Supreme Court has reduced the nine-year sentence of a woman charged with burning a man alive, to four years.
    2. The woman decided to set fire to a customer of the bar where she worked.
    3. The customer she fired had raped her daughter a few months ago.
    4. The customer she fired was on temporary release from prison at the time of the incident and decided to go to the woman's workplace to make fun of her.

- Dilemma 2:

    1. A mother lets her two-year-old baby die.
    2. The baby was admitted to hospital in critical condition. He urgently needed to undergo surgery as well as a blood transfusion.

3. The mother did not authorize the blood transfusion since, being a Jehovah's Witness, her religion prohibits it.
4. Theere would have been a chance of saving the baby had the transfusion been carried out.

Now, think if the "Affect Heuristic" has had some impact on your "accept" decision or not.

Deliverables:

- **Exercises:**
  - **Session 1**: 6 and 7.
  - **Session 2:** 2 and 4 (only the part of the Sprinkler problem).
- **Delivery method:** Virtual Campus
- **Deadline:** Sunday, 6^{th} April